

Build a Pub Sub to Cloud Run Event Pipeline

105 minutes · Mapped criteria A3, A4, K4

Purpose and work product

Deploy a small event receiver, publish mock orders and diagnose retry behaviour without losing messages.

Submit: Cloud Run revision, Pub/Sub subscription and processed event log

Safety, cost and privacy

Use only the assigned training project. Confirm the active account, project and region before every mutation. Never place credentials, payment details, access tokens or private keys in screenshots or submitted files. Stop billable resources and verify the final inventory.

Files in this folder

File	Purpose
app.py	Mock data, script or configuration used by this activity
requirements.txt	Mock data, script or configuration used by this activity
orders.jsonl	Mock data, script or configuration used by this activity

Detailed procedure

Step 1

Review app.py, requirements.txt and orders.jsonl. Confirm the receiver logs message ID and payload but no secret.

Step 2

Create the orders topic and a subscription, then publish one test message manually.

Step 3

Build and deploy ace-api to Cloud Run in asia-southeast1 with authenticated ingress.

Step 4

Create a Pub/Sub push subscription or Eventarc trigger using the Cloud Run service identity and documented invoker permissions.

Step 5

Publish both mock orders and query Cloud Run revision logs for the two message IDs.

Step 6

Send the same message ID twice in the mock harness and record the duplicate-delivery risk.

Step 7

Add or describe an idempotency control keyed on order_id/message ID.

Step 8

Introduce a harmless 500 response for one test event and observe retry/backlog behaviour.

Step 9

Restore the 204 response and verify the backlog returns to zero.

Step 10

Save the revision name, trigger/subscription configuration and processed event evidence.

Step 11

Delete the service, topic and subscription unless the trainer asks you to retain them.

Verification record

Record expected and observed output. A successful command without the correct project, identity, scope or evidence is not an accepted result.

Check	Expected evidence
Identity and scope	Active account, project ID, region or zone
Configuration	Resource description or policy output
Runtime result	Test output with UTC timestamp
Failure test	One controlled negative test and diagnosis
Cleanup	Final inventory and stopped/deleted billable resources

Troubleshooting sequence

Check the active identity and project first, then API enablement and IAM, then resource state, networking and quotas. Preserve the failed output before making a correction. If the issue remains, escalate with the exact command, timestamp, expected result, observed result and evidence files.

Cleanup

Delete or stop resources as instructed. Re-open Billing and the relevant service inventory, then record the remaining resource names. Budget alerts do not stop charges.